

Name: _____



Recognising special number sequences

Task A: Arithmetic and geometric sequences

1) Sort these sequences into the correct column of the table.

arithmetic	geometric	neither

- a) 1, 2, 3, 4, 5, ...
- b) 3, 9, 27, 81, ...
- c) 0.5, 1, 2, 4, ...
- d) 1, 1, 2, 4, 7, ...
- e) 1, 2, 4, 7, 11, ...
- f) -2, 4, -8, 16, ...
- g) -1, 5, 11, 17, 23, ...
- h) 18, 35, 52, 68, ...
- i) 36, 12, -12, -36 ...
- j) 13.5, 9, 6, 4, ...
- k) 12, 6, 3, 0, ...

2) Sort these sequences into the correct column of the table.

arithmetic	geometric	neither

- a) $\frac{3}{2}, 1, \frac{1}{2}, 0, \dots$
- b) $\frac{3}{8}, \frac{3}{4}, \frac{3}{2}, 3, \dots$
- c) $\frac{1}{6}, \frac{5}{3}, \frac{50}{3}, \frac{500}{3}, \dots$
- d) $\frac{3}{5}, \frac{3}{10}, \frac{3}{15}, \frac{3}{20}, \dots$
- e) $-3, 1, -\frac{1}{3}, \frac{1}{9}, \dots$
- f) $-\frac{8}{5}, 1, \frac{18}{5}, 6, \dots$
- g) $-\frac{4}{5}, -\frac{2}{3}, -\frac{8}{15}, -\frac{2}{5}, \dots$

3) A sequence starts $2a, 6a, \dots$

a) If the sequence was arithmetic, what would the next three terms be?

b) If the sequence was geometric, what would the next three terms be?

c) Continue this sequence in another way, describe the rule you have used.

Task B: Exploring other sequences

1) By working out the pattern, find the next two terms in each sequence.

- a) 2, 8, 32, 128, ...
- b) 2, 8, 14, 20, ...
- c) 2, 8, 15, 23, ...
- d) 2, 8, 16, 26, ...
- e) 2, 8, 10, 18, ...
- f) 2, 8, 17, 29, ...
- g) 8, 2, 10, 12, ...
- h) 8, 2, -4, -10, ...
- i) 3, 5, 11, 21, ...
- j) 4, 6, 12, 22, ...
- k) 6, 10, 22, 42, ...
- l) -3, -5, -8, -13, ...
- m) -3, -5, -7, -9, ...
- n) 2, -6, 18, -54, ...
- o) 1, -7, -10, -8, ...

2) Aisha, Izzy and Lucas have all created a sequence starting with the terms

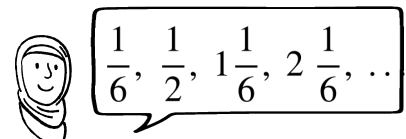
$$\frac{1}{6}, \frac{1}{2}, \dots$$

For each question justify your answer.

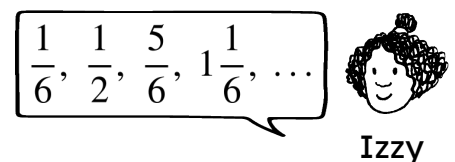
a) Who has created an arithmetic sequence?

b) Who has created a geometric sequence?

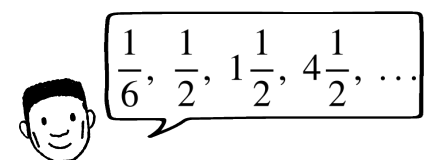
c) Who has created a quadratic sequence?



Aisha



Izzy



Lucas